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A Project Report on

Preparation of Multigrain Bread

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1. Introduction

1.1 Overview of the project

This project focuses on the preparation of **Multigrain Bread**, a nutritious bakery product that combines several grains such as wheat, oats, millet, and barley. The prototype demonstrates the chemical and biological processes involved in bread making — primarily **fermentation** using yeast and **baking** as a heat treatment step.

1.2 Purpose of the prototype production

The main objective of this prototype is to understand the **chemical and process engineering principles** used in food manufacturing, such as mixing, fermentation, and heat transfer. It provides practical exposure to batch manufacturing, quality testing, and process optimization at a small scale.

1.3 Scope of the report

This report includes the complete process description, materials and equipment used, step-by-step experimental procedure, process parameters, testing methods, cost analysis, and conclusions. It also serves as documentation of hands-on experience gained in the Chemical Prototyping module.

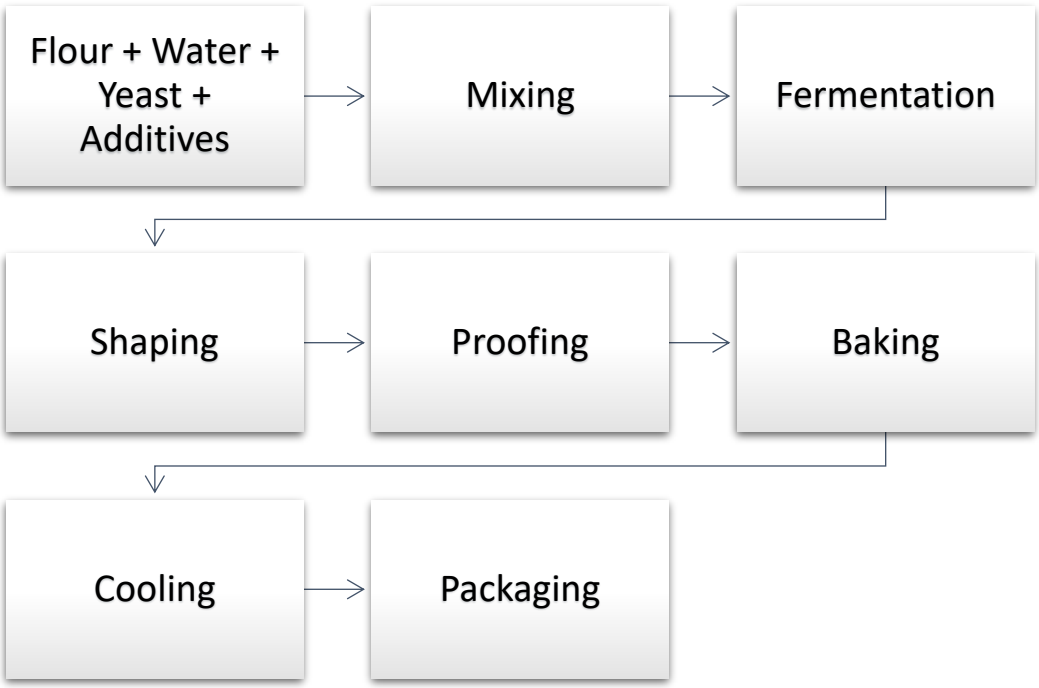
2. Process Description

2.1 Summary of the manufacturing process

The process involves the following key stages:

1. Mixing of ingredients
2. Fermentation of dough using yeast
3. Shaping and proofing
4. Baking at controlled temperature
5. Cooling and packaging

2.2 Flowchart / Process Diagram



2.3 Key process steps and sequence

- Ingredient preparation and weighing
- Dough mixing and kneading
- Fermentation (biochemical reaction)
- Heat treatment during baking
- Post-processing and packaging

3. Raw Materials and Chemicals

3.1 List of raw materials and chemicals used

Sr. No.	Material	Function	Quantity (approx.)
1	Multigrain flour (wheat, oats, millet, etc.)	Base ingredient	250 g
2	Yeast	Fermentation agent	10 g
3	Water	Solvent & hydration	150 mL
4	Sugar	Nutrient for yeast	20 g
5	Salt	Flavor enhancer	5 g
6	Vegetable oil	Softening agent	20 mL

3.2 Sources and suppliers

- All materials were food-grade, purchased from local grocery suppliers or the college lab store.

3.3 Material specifications and safety considerations

- Ensure yeast is fresh and active.
- Maintain hygiene to avoid contamination.
- Handle all materials with clean utensils.

4. Equipment and Instrumentation

4.1 Description of equipment used

1. Digital weighing balance
2. Measuring cylinder
3. Mixing bowl and Spoon
4. Oven (Pre-Heat to 180°C)
5. Thermometer
6. Baking tray

4.2 Specifications of machinery

- **Oven:** Electric type, temperature range 50–200°C
- **Balance:** Accuracy ±1 g
- **Thermometer:** 0–180°C range

4.3 Instrumentation and control systems

- Manual temperature control and timer used for baking and fermentation monitoring.

5. Experimental Procedure / Prototype Fabrication

5.1 Step-by-step procedure

1. **Weighing:** Accurately measure all ingredients.
2. **Mixing:** Combine flour, yeast, sugar, and salt. Add water and oil gradually.
3. **Kneading:** Mix until dough becomes smooth and elastic (10–12 minutes).
4. **Fermentation:** Keep dough in a warm environment (30–35 °C) for 1.5–2 hours.
5. **Shaping:** After fermentation, lightly knead and shape the dough into a loaf.
6. **Proofing:** Allow dough to rest for 30–45 minutes for secondary rise.
7. **Baking:** Bake in a preheated oven at 180 °C for 25–30 minutes.
8. **Cooling:** Remove loaf and cool on a rack for 30 minutes.
9. **Packaging:** Wrap in clean film or bag after cooling.

5.2 Conditions maintained

- Fermentation Temperature: **30–35 °C**
- Fermentation Time: **2 hours**
- Baking Temperature: **180 °C**
- Baking Time: **30 minutes**

5.3 Safety measures and protocols

- Use gloves while handling hot trays.
- Keep workspace sanitized.
- Do not overheat oven or use metallic utensils inside.

6. Process Parameters and Optimization

6.1 Key process variables

- Fermentation temperature
- Fermentation time
- Hydration ratio (flour-to-water ratio)
- Baking temperature and duration

7. Quality Control and Testing

7.1 Testing methods and standards used

- **Visual inspection:** Color, crust, and shape
- **Texture test:** Softness and elasticity
- **Taste and aroma:** Sensory evaluation
- **Moisture content:** Estimated via weight loss

7.2 Results of prototype evaluation

- Bread was uniformly baked, had a soft texture, and pleasant aroma. No cracks or dense areas were observed.

7.3 Comparison with expected specifications

Parameter	Expected	Obtained
Color	Golden brown	Golden brown
Texture	Soft and fluffy	Soft and uniform
Taste	Slightly sweet	Acceptable
Weight loss	10–12%	10%

8. Yield and Efficiency Analysis

8.1 Mass balance calculations

Total input:
Flour (250 g) + Yeast (10 g) + Water (150 g) + Sugar (20 g) + Salt (5 g) + Oil (20 g) = **455 g**

Output (after baking): ~400 g

Yield percentage

$$\text{Yield} = \frac{455}{400} \times 100 = 113.75\%$$

8.2 Efficiency analysis

Moisture loss of approximately 10% observed, indicating efficient baking without over-drying.

9. Challenges and Troubleshooting

9.1 Issues encountered

- Uneven fermentation in early trials.
- Slightly dense texture when hydration was low.
- Overbrowning in some batches.

9.2 Solutions implemented

- Controlled temperature and humidity during fermentation.
- Increased water ratio from 55% to 60%.
- Optimized baking temperature to 200 °C.

Lessons learned

Maintaining precise fermentation time and temperature is crucial for achieving ideal bread texture and taste.

10. Environmental and Safety Considerations

10.1 Waste generation and disposal

- Minimal waste generated (mainly dough residue).
- Organic waste disposed in compost bin.
- Cleaning water discharged through regular drainage.

10.2 Environmental impact assessment

The process uses locally available raw materials and is environmentally safe. Baking energy consumption is moderate.

10.3 Safety hazards and risk mitigation

- Used insulated gloves for hot trays.
- Avoided direct contact with electrical oven components.
- Maintained cleanliness to prevent biological contamination.

11. Cost Analysis

Item	Quantity	Cost (₹)
Multigrain flour	**1Kg**	**71/-**
Yeast	**25 g**	**40/-**
Sugar, Salt, Oil	**-**	**40/-**
Electricity / Fuel	**—**	**10/-**
Total Cost		**₹161/-**

11.1 Equipment and operational costs

Since lab equipment was institutional, depreciation and maintenance were not included.
Operational energy cost estimated at ₹5 per batch.

11.2 Economic feasibility assessment

The process is low-cost and feasible for small-scale or cottage food production units.

12. Conclusions and Recommendations

12.1 Summary of findings

- The **multigrain bread prototype** was successfully developed with good texture, appearance, and yield.
  Optimal process parameters were:
  - Fermentation: 2 hours at 30–35 °C
  - Baking: 200 °C for 30 minutes

12.2 Recommendations for improvement

- Automate temperature control for consistent fermentation.
- Test different grain combinations for better nutrition.
- Conduct shelf-life and microbial testing for extended storage.

12.3 Next steps in product development

- Scale up the batch to pilot scale and test packaging materials for moisture retention and freshness.

13. References

Video link : https://youtu.be/q_dSCQhIf3Y?si=tRGTjG_tzHpeKsl1

15. Photos and YouTube Video Link

YouTube video link paper model :

https://youtube.com/shorts/_B4Jj276GVo?si=XE-fMlGEVQJeMr7e

Link of Cooking process video:
https://drive.google.com/file/d/1MUTJPtsR6B7DxspCo7lvXZCkDGUIz-Th/view?usp=drivesdk

Photos:








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